

RAJEEV INSTITUTE OF TECHNOLOGY

HASSAN-573201

CLOUD COMPUTING LABORATORY

(BCS601)

As per VTU Syllabus/scheme for 6th Semester

Prepared By:

Prof .Nanditha S R
Asst Professor



**DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING**



VISION & MISSION OF THE INSTITUTE

VISION:

- ❖ To be an academic institution in vibrant social & economic environment, striving continuously for excellence in education, research and technological service to the society.

MISSION:

- ❖ To achieve academic excellence in engineering and management through dedication to duty, offering state of the art education and faith in human values.
- ❖ To create and endure a community of learning among students, develop outstanding professionals with high ethical standards.
- ❖ To provide academic ambience conducive to the development, needs and growth of society and the industry.

VISION & MISSION OF THE DEPARTMENT

VISION:

- ❖ To become an exemplary center for engineering education and research in the frontier areas of Computer Science and Engineering.

MISSION:

- ❖ Render quality education through best teaching learning process and modern educational tools to enable students for good careers.
- ❖ Impart essential skills to make students industry ready and future professional with social concern.
- ❖ Provide facilities and expertise in recent computer technology to promote research and innovations.



PROGRAM SPECIFIC OUTCOMES (PSO'S)

The graduates of Computer Science & engineering program of Rajeev Institute of Technology should be able to attain the following at the time of graduation.

PSO1: To apply software engineering principles and practices to design, develop and implement software solutions.

PSO2: To use recent software tools to solve and analyze a given problem.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

The program educational objectives are the statements that describe the expected achievements of graduates within first few years of their graduation from the program. The program educational objectives of Bachelor of Computer Science & Engineering at Rajeev Institute of Technology can be broadly defined as,

PEO1: Graduates will become software developers in diverse domains and/or pursue higher studies.

PEO2: Graduates will possess an agility to adapt to challenges in their profession.

PEO3: Graduates exhibit leadership qualities and will be a team player.

PEO4: Graduates will be able to communicate effectively and cultivate ethics.



PROGRAM OUTCOMES

Graduation students of Bachelor of Computer Science and Engineering program at Rajeev Institute of Technology will attain the following program outcomes in the field of Computer Science and Engineering.

PO1-Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2-Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3-Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4-Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5-Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling to complex engineering activities, with an understanding of the limitations.

PO6-The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7-Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8-Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9-Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.



PO10-Communication: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11-Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environment.

PO12-Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



Practical Components

Experiment-1

Exploring AWS CloudShell and the AWS Cloud9 IDE

1.1 Apparatus Required: AWS Account

1.2 Pre-Requisite: Basic Linux knowledge

1.3 Introduction:

AWS Cloudshell - AWS CloudShell is a browser-based shell that makes it easier to securely manage, explore, and interact with your AWS resources. AWS Cloud9 IDE - AWS Cloud9 is a cloud-based integrated development environment (IDE) that lets you write, run, and debug your code with just a browser. It includes a code editor, debugger, and terminal.

1.4 Procedure:

1. Login to the AWS console using AWS account Credentials.



Sign in

☒ **Root user**

Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☐ **IAM user**

User within an account that performs daily tasks. [Learn more](#)

Root user email address

username@example.com

Next

By continuing, you agree to the [AWS Customer Agreement](#) or other agreement for AWS services, and the [Privacy Notice](#). This site uses essential cookies. See our [Cookie Notice](#) for more information.

____ New to AWS? ____

Create a new AWS account

**Amazon Nova, new
foundation models that
deliver frontier intelligence
and industry leading
price performance**

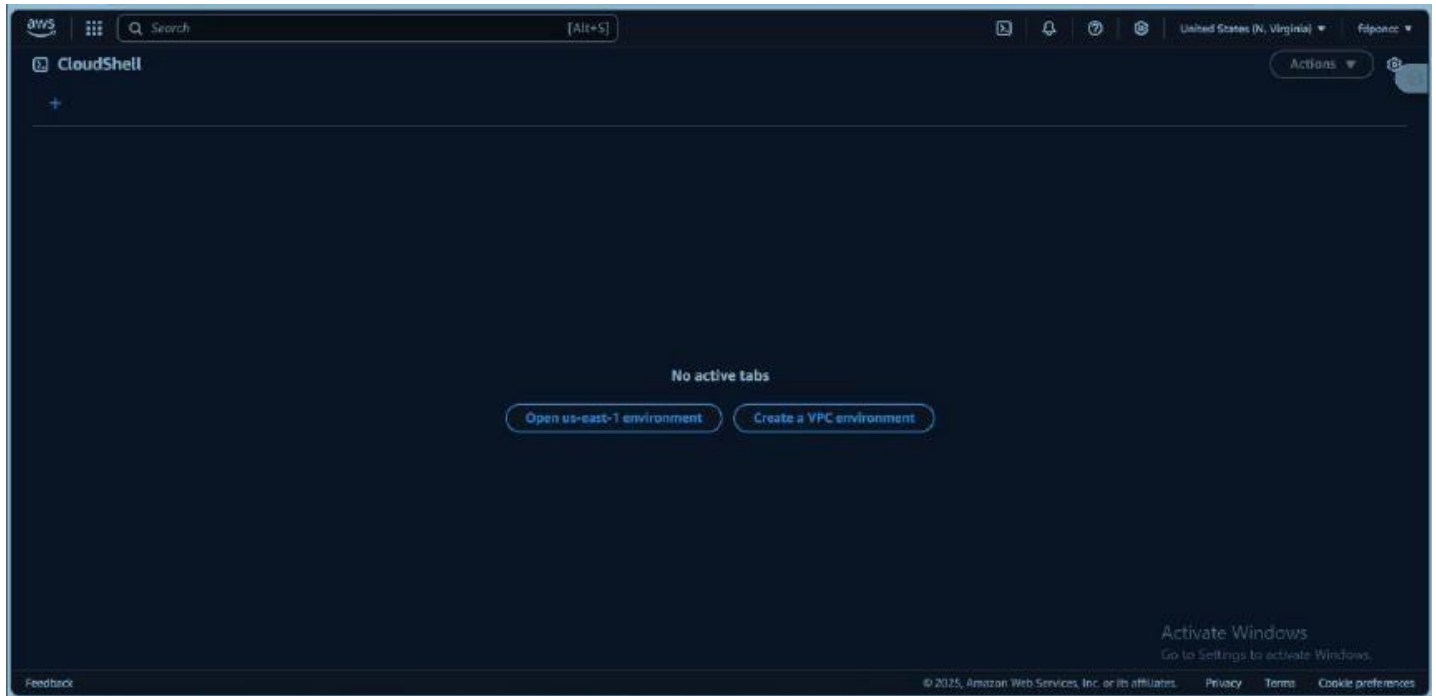
[Learn more >](#)



2. Select CLOUDSHELL



3. CREATE USING AN INSTANCE



4. cloud shell terminal



5. Execute basic shell commands

Shell Prompt

The prompt, \$, which is called the command prompt, is issued by the shell. While the prompt is displayed, you can type a command.

Shell reads your input after you press Enter. It determines the command you want executed by looking at the first word of your input. A word is an unbroken set of characters. Spaces and tabs separate words.

Following is a simple example of the command,

\$date	# Displays the current date
\$pwd	# Displays the present working directory.



\$ls # Lists the contents of the current directory.
\$echo # Prints a string of text, or value of a variable to the terminal.

Create and Execute Bash scripts

```
#!/bin/bash  
echo "Today is " `date`
```

Line #1: The shebang (#!/bin/bash) points toward the bash shell path.

Line #2: The echo command is displaying the current date and time on the terminal. Note that the date is in backticks.

Executing the bash script

To make the script executable, assign execution rights to your user using this command:

```
chmod 777 hello.sh
```

You can run the script using any of the mentioned methods:

- sh hello.sh
- bash hello.sh
- ./hello.sh

```
aws | Search [Alt+S] | Europe (Stockholm) | SJP_RIT_HSN | Actions |  
CloudShell  
eu-north-1 +  
~ $ date  
Fri Apr 4 07:41:13 AM UTC 2025  
~ $ pwd  
/home/cloudshell-user  
~ $ mkdir RIT_CC_601  
~ $ ls  
add.py hello.js hello.js RIT_CC_601 test.php test.py  
~ $ cd RIT_CC_601  
RIT_CC_601 $ pwd  
/home/cloudshell-user/RIT_CC_601  
RIT_CC_601 $ ls  
RIT_CC_601 $ echo "Hello bash"  
Hello bash  
RIT_CC_601 $ vi hello.sh  
RIT_CC_601 $ chmod 777 hello.sh  
RIT_CC_601 $ ./hello.sh  
Today is Fri Apr 4 07:52:52 AM UTC 2025  
RIT_CC_601 $
```



6.UPLOAD A FILE

The screenshot shows the AWS CloudShell interface. The terminal window displays the following commands and output:

```
~ $ ls
add.py hello1.js hello.js RIT_CC_601 test.php test.py
~ $ vi add.py
~ $ cd RIT_CC_601
RIT_CC_601 $ vi hello.py
RIT_CC_601 $ python3 hello.py
Hello, World!
RIT_CC_601 $ python3 Add.py 3 4
python3: can't open file '/home/cloudshell-user/RIT_CC_601/Add.py': [Errno 2] No such file or directory
RIT_CC_601 $ ls -l
total 8
-rwxr-xr-x. 1 cloudshell-user cloudshell-user 23 Apr  4 09:22 hello.py
-rwxr-xr-x. 1 cloudshell-user cloudshell-user 36 Apr  4 07:52 hello.sh
RIT_CC_601 $ cls
bash: cls: command not found
RIT_CC_601 $ clc
bash: clc: command not found
RIT_CC_601 $ cd ..
~ $ vi hello.js
~ $ cd RIT_CC_601
RIT_CC_601 $ vi hello.js
RIT_CC_601 $ cat hello.js
console.log('This example is different!');
console.log('The result is displayed in the Command Line Interface');
console.log('Hello, World!');
RIT_CC_601 $ node hello.js
This example is different!
The result is displayed in the Command Line Interface
Hello, World!
```

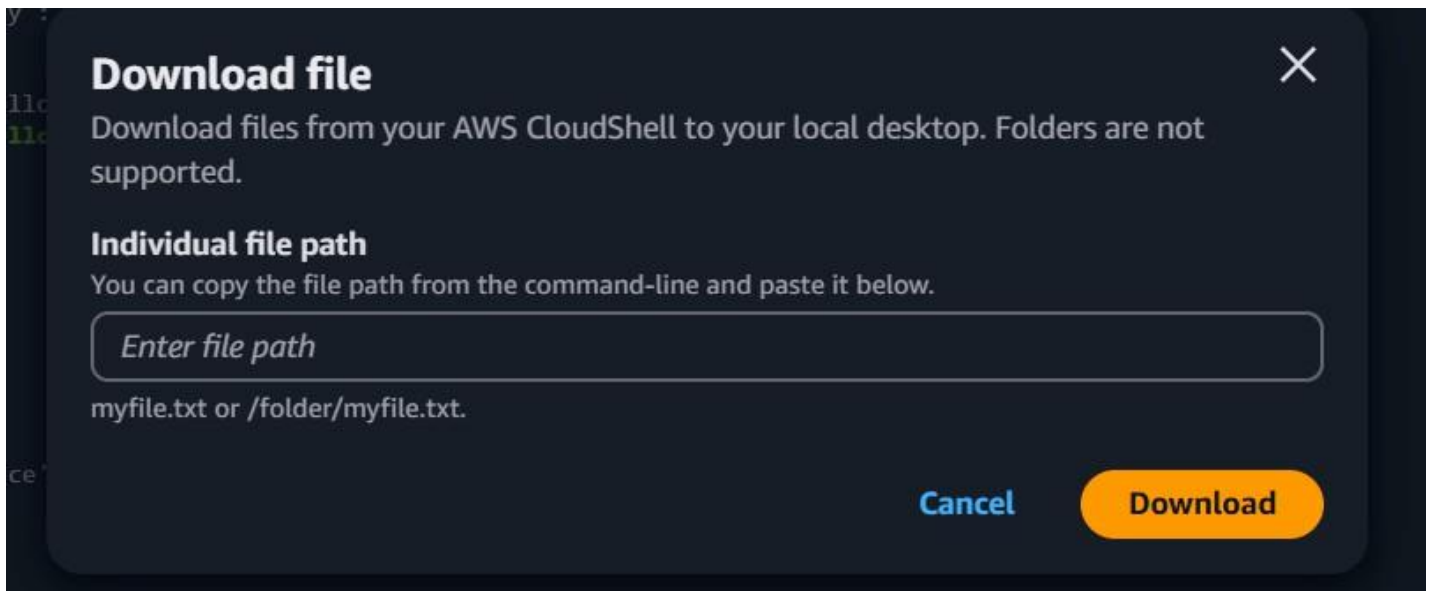
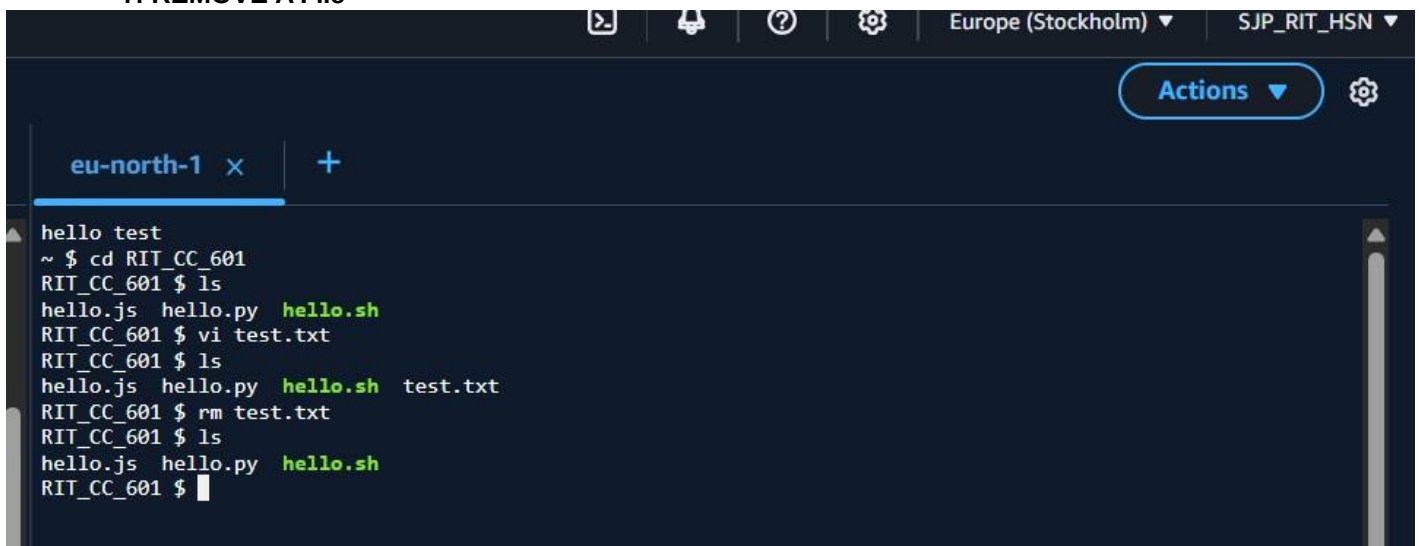
Two green notification banners indicate successful file uploads:

- File upload successful
hello.txt.txt was successfully uploaded to the following directory: /home/cloudshell-user.
- File upload successful
hello.txt.txt was successfully uploaded to the following directory: /home/cloudshell-user.

5. SPLIT INTO MULTIPLE ENVIRONMENTS

The screenshot shows the AWS CloudShell interface with two tabs open, both named "eu-north-1". The left tab shows the same commands and output as the previous screenshot. The right tab shows the following commands and output:

```
~ $ cd RIT_CC_601
RIT_CC_601 $ ls
hello.js hello.py hello.sh
RIT_CC_601 $
```

6. DOWNLOAD A FILE**7. REMOVE A File**



Working with An EC2 instance in AWS, is a virtual server within the Amazon Elastic Compute Cloud (EC2), a scalable cloud computing platform, where you can run applications on the AWS cloud.

1. login



Sign in

☒ **Root user**

Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☐ **IAM user**

User within an account that performs daily tasks. [Learn more](#)

Root user email address

username@example.com

Next

By continuing, you agree to the [AWS Customer Agreement](#) or other agreement for AWS services, and the [Privacy Notice](#). This site uses essential cookies. See our [Cookie Notice](#) for more information.

New to AWS?

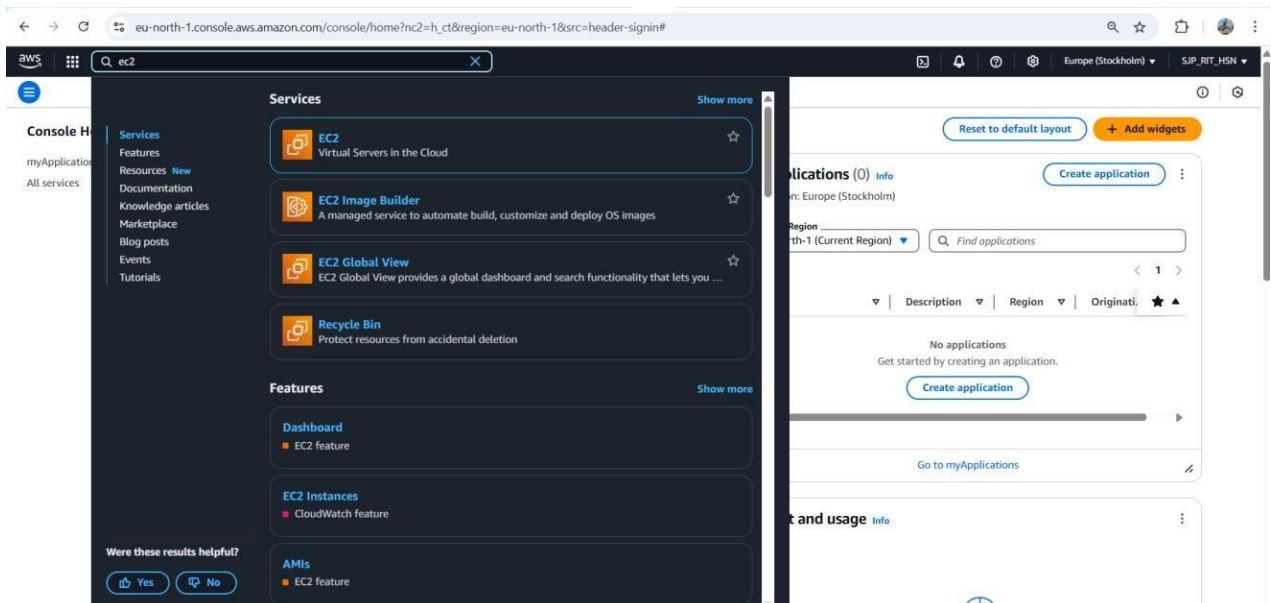
Create a new AWS account

Business Support Limited Time Offer

Expert technical assistance
when you need it most

[Learn more ›](#)

2. create EC2





3. Goto EC2 Dashboard

The screenshot shows the AWS Management Console for the EC2 service in the eu-north-1 region. The dashboard provides an overview of EC2 resources and offers a 'Launch instance' button. The 'Resources' section lists various EC2 components like Instances, Auto Scaling Groups, and Capacity Reservations. The 'Launch instance' section includes a 'Launch instance' button and a 'Migrate a server' button. The 'Service health' section shows the status of the EC2 service, and the 'Zones' section lists available availability zones. The 'Account attributes' section displays the default VPC and settings.

Resources

You are using the following Amazon EC2 resources in the Europe (Stockholm) Region:

Resource	Count
Instances (running)	0
Dedicated Hosts	0
Key pairs	0
Security groups	1
Auto Scaling Groups	0
Elastic IPs	0
Load balancers	0
Snapshots	0
Capacity Reservations	0
Instances	0
Placement groups	0
Volumes	0

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

[Launch instance](#) [Migrate a server](#)

Note: Your instances will launch in the Europe (Stockholm) Region

Instance alarms

[View in CloudWatch](#)

0 in alarm 0 OK 0 insufficient data

Service health

[AWS Health Dashboard](#)

Region: Europe (Stockholm)

Status: This service is operating normally.

Zones

Zone name	Zone ID
eu-north-1a	eun1-az1
eu-north-1b	eun1-az2
eu-north-1c	eun1-az3

Account attributes

[Default VPC](#)

vpc-099c6e2ad99b2b308

Settings

- Data protection and security
- Allowed AMIs
- Zones
- EC2 Serial Console
- Default credit specification
- EC2 console preferences

4. To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

The screenshot shows the 'Launch an instance' wizard in the AWS Management Console. It provides a step-by-step guide to launching an EC2 instance. The 'Name and tags' section allows users to enter a name and add tags. The 'Application and OS Images (Amazon Machine Image)' section provides a search bar to find the desired AMI. The 'Quick Start' section offers pre-configured templates for various operating systems like Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, and Debian. The 'Summary' section displays the configuration details, including the number of instances, software image, virtual server type, firewall, and storage. A 'Free tier' notification is also present, indicating that the user is eligible for free usage of certain instance types.

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name: [Add additional tags](#)

Application and OS Images (Amazon Machine Image)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.7.2...[read more](#)
ami-0274f4b62b6ae3bd5

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

[Cancel](#) [Launch instance](#)



5. Launch an instance

[Alt+S]

[EC2](#) > [Instances](#) > Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

 [Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-0c1ac8a41498c1a9c (64-bit (x86)) / ami-09fdd0b7882a4ec7b (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture	AMI ID	Publish Date	Username	
64-bit (x86)	ami-0c1ac8a41498c1a9c	2025-03-05	ubuntu	Verified provider
64-bit (x86)				
64-bit (Arm)				

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour On-Demand RHEL base pricing: 0.0396 USD per Hour

On-Demand SUSE base pricing: 0.0108 USD per Hour On-Demand Linux base pricing: 0.0108 USD per Hour

On-Demand Windows base pricing: 0.02 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)



▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select

[Create new key pair](#)

Q |

Proceed without a key pair (Not recommended)

Default value

[Edit](#)

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

SJP_RIT

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒

RSA

RSA encrypted private and public key pair

☐

ED25519

ED25519 encrypted private and public key pair

Private key file format

☒

.pem

For use with OpenSSH

☐

.ppk

For use with PuTTY



When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

[Cancel](#)

[Create key pair](#)

SJP_RIT.pem file created

-----BEGIN RSA PRIVATE KEY-----

```
MIIEpAIBAAKCAQEAPFrNmCBUqo3kJyCRvil0VmQ1uqF4omxR+zpY+VayTWmbk9k
0/R/PITZhqcoDIGGrGzqhMP9upf8dNMDEBZ02y7rx0nZxlnedb+Ezhpn6I0jW+Ln
oZtIsY8yoM/LcUQH3Zvke1mofUNKuP+rj/xYSxRdW9nq8l4jkEs0e/NqJCDAXyZU
Q02t5l5F9jjFvGIj0Iygn81lZ6Ef6i3bptyX0surwEg7o4eV6s2csCSYY7sopYys
88hK0Rzgn9RpAKt36GITMXrMi6onwgKHj6dAd4FCIFLPykBuitlnwyKXFLvSiG8N
1RRckda+8X/onUnbQ8jcIZ+TPN+URTt5/1/IzwIDAQABAoIBAFXGMO4IQ/f6H9lZ
```

```
tzSwiUfa4YMNs0CwqEhT7/Kffcf/kkHC5PTTsr+BDRDSvnEy/X+Kmhjh3G8xG78L
cZJCHL5P6QJNEX2whi5UfTSAZBqsSyuy+YPEofOwsmk92ngmspg0TgObwowGE9IO
c3ZtQqKBgQCugVo+GAX90M7F74vViSywhdrglgOeZsLEIfmitGhR5n+AIpsff85p
zOnmY5WVx+PZikgYvVWz0tZK1fuf51QqaqJhTK20M1yyGQqk+/lyv/XgJxtQNBBN
/x+PLiKGgdFGqYTXIueHGNMMGDCmv1TNnqmNodqiT9X8F3h+bRLdMw==
```

-----END RSA PRIVATE KEY-----



▼ Network settings [Info](#)

[Edit](#)

Network [Info](#)

vpc-099c6e2ad99b2b308

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- ☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

- ☒ Allow SSH traffic from Anywhere
0.0.0.0/0
Helps you connect to your instance
- ☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
- ☐ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

▼ Configure storage [Info](#)

[Advanced](#)

1x GiB Root volume, 3000 IOPS, Not encrypted

ⓘ Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

The selected AMI contains more instance store volumes than the instance allows. Only the first 0 instance store volumes from the AMI will be accessible from the instance

🕒 Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

[Edit](#)

► Advanced details [Info](#)



Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64 noble image
ami-0c1ac8a41498c1a9c

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

[Cancel](#)

[Launch instance](#)

[Preview code](#)

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

Search [Alt+S] Europe (Stockholm) SJP_RIT_HSN

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0537e4e677258c7ec)

[Launch log](#)

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

1 2 3 4 5 6

Create billing and free tier usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

[Create billing alerts](#)

Connect to your instance

Once your instance is running, log into it from your local computer.

[Connect to instance](#)

[Learn more](#)

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

[Connect an RDS database](#)

[Create a new RDS database](#)

[Learn more](#)

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

[Create EBS snapshot policy](#)



6. CONNECT TO INSTANCE

EC2

Dashboard

EC2 Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Instances (1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

1

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☐	SJP_RIT_HSN1	i-0537e4e677258c7ec	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-13-61-33-146.eu-n...	13.61.2

Select an instance

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:

Search

[Alt+S]

Europe (Stockholm)

SJP_RIT_HSN

EC2

Instances

Instances (1/1) Info

Last updated 1 minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

1

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☒	SJP_RIT_HSN1	i-0537e4e677258c7ec	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-13-61-33-146.eu-n...	13.61.2

i-0537e4e677258c7ec (SJP_RIT_HSN1)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary Info

Instance ID

i-0537e4e677258c7ec

IPv6 address

-

Public IPv4 address

13.61.33.146 | open address

Instance state

Running

Private IPv4 addresses

172.31.46.92

Public IPv4 DNS

ec2-13-61-33-146.eu-north-1.compute.amazonaws.com | open address

Hostname type

IP name: ip-172-31-46-92.eu-north-1.compute.internal

Private IP DNS name (IPv4 only)

ip-172-31-46-92.eu-north-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-099c6e2ad99b2b308

Answer private resource DNS name

IPv4 (A)

13.61.33.146 [Public IP]

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more



1. List the instance Summary of i-0537e4e677258c7ec (SJP_RIT_HSN1)

Instance ID	IPv6 address
Public IPv4 address	Public IPv4 DNS
Private IPv4 addresses	Hostname type
Instance state	Private IP DNS name (IPv4 only)
Platform details	Launch time
Key pair assigned at launch	Owner
Virtualization type	Number of vCPUs

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#ConnectToInstance:instanceId=i-0537e4e677258c7ec

Search [Alt+S]

Europe (Stockholm) SJP_RIT_HSN

EC2 > Instances > i-0537e4e677258c7ec > Connect to instance

Connect to instance

Connect to your instance i-0537e4e677258c7ec (SJP_RIT_HSN1) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID

i-0537e4e677258c7ec (SJP_RIT_HSN1)

Connection Type

☒ Connect using EC2 Instance Connect

☐ Connect using EC2 Instance Connect Endpoint

☒ Public IPv4 address

☐ IPv6 address

13.61.33.146

Username

ubuntu

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect

aws

Search [Alt+S]

Europe (Stockholm) SJP_RIT_HSN

Establishing Connection ...

i-0537e4e677258c7ec (SJP_RIT_HSN1)

PublicIPs: 13.61.33.146 PrivateIPs: 172.31.46.92

CloudShell Feedback © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



```
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Mon Apr 7 07:09:50 UTC 2025

System load: 0.15          Temperature: -273.1 C
Usage of /: 25.0% of 6.71GB Processes: 107
Memory usage: 22%         Users logged in: 0
Swap usage: 0%            IPv4 address for ens5: 172.31.46.92

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-46-92:~$
```

i-0537e4e677258c7ec (SJP_RIT_HSN1)

PublicIPs: 13.61.33.146 PrivateIPs: 172.31.46.92

```
Swap usage: 0%            IPv4 address for ens5: 172.31.46.92

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

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ubuntu@ip-172-31-46-92:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-46-92:~$ date
Mon Apr 7 07:12:00 UTC 2025
ubuntu@ip-172-31-46-92:~$ ps
  PID TTY          TIME CMD
 1528 pts/0    00:00:00 bash
 1577 pts/0    00:00:00 ps
ubuntu@ip-172-31-46-92:~$
```

i-0537e4e677258c7ec (SJP_RIT_HSN1)

PublicIPs: 13.61.33.146 PrivateIPs: 172.31.46.92



aws [Search] [Alt+S] Europe (Stockholm) SJP_RIT_HSN

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PID TTY          TIME CMD
1528 pts/0    00:00:00 bash
1577 pts/0    00:00:00 ps
ubuntu@ip-172-31-46-92:~$ vi hello.sh
ubuntu@ip-172-31-46-92:~$ ls -l hello.sh
-rw-rw-r-- 1 ubuntu ubuntu 77 Apr 7 07:13 hello.sh
ubuntu@ip-172-31-46-92:~$ chmod 777 hello.sh
ubuntu@ip-172-31-46-92:~$ ls -l
total 4
-rwxrwxrwx 1 ubuntu ubuntu 77 Apr 7 07:13 hello.sh
ubuntu@ip-172-31-46-92:~$ ./hello.sh
Today is Mon Apr 7 07:14:49 UTC 2025
ubuntu@ip-172-31-46-92:~$
```

i-0537e4e677258c7ec (SJP_RIT_HSN1)
PublicIPs: 13.61.33.146 PrivateIPs: 172.31.46.92

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:

aws [Search] [Alt+S] Europe (Stockholm) SJP_RIT_HSN

EC2 > Instances

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm state
SJP_RIT_HSN1	i-0537e4e677258c7ec	Running	t3.micro	3/3 checks passed	View alarm

Stop instance
Start instance
Reboot instance
Hibernate instance
Terminate (delete) instance

i-0537e4e677258c7ec (SJP_RIT_HSN1)

2. stop instance

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:

aws [Search] [Alt+S] Europe (Stockholm) SJP_RIT_HSN

EC2 > Instances

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Stop instance
Start instance
Reboot instance
Hibernate instance
Terminate (delete) instance

i-0537e4e677258c7ec (SJP_RIT_HSN1)



Stop instance

Stopping your instance allows you to reduce costs, modify settings, and troubleshoot problems.

Instance ID

Stop protection

[i-0537e4e677258c7ec \(SJP_RIT_HSN1\)](#)

Off (Can stop instance)



You will be billed for associated resources

After you stop the instance, you are no longer charged usage or data transfer fees for it. However, you will still be billed for associated Elastic IP addresses and EBS volumes.

Associated resources

You will continue to incur charges for these resources while the instance is stopped

Cancel

Stop

aws [Search] [Alt+S] Europe (Stockholm) SJP_RIT_HSN

EC2 > Instances

EC2

- Dashboard
- EC2 Global View
- Events

▼ Instances

- Instances
- Instance Types
- Launch Templates

Successfully initiated stopping of i-0537e4e677258c7ec

Instances (1/1) Info

Last updated less than a minute ago

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☒	SJP_RIT_HSN1	i-0537e4e677258c7ec	Stopping	t3.micro	3/3 checks passed	View alarms	eu-north-1b	ec2-13-61-33-146.eu-n...	13.61.2

3. delete instance

aws [Search] [Alt+S] Europe (Stockholm) SJP_RIT_HSN

EC2 > Instances

EC2

- Dashboard
- EC2 Global View
- Events

▼ Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances

Successfully initiated stopping of i-0537e4e677258c7ec

Instances (1/1) Info

Last updated less than a minute ago

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☒	SJP_RIT_HSN1	i-0537e4e677258c7ec	Stopped	t3.micro	3/3 checks passed	View alarm			

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate (delete) instance



Terminate (delete) instance

⚠ On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.

Are you sure you want to terminate these instances?

Instance ID

Termination protection

i-0537e4e677258c7ec (SJP_RIT_HSN1)

Disabled

To confirm that you want to delete the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.

Cancel

Terminate (delete)

aws [Search] [Alt+S] Europe (Stockholm) SJP_RIT_HSN

EC2 > Instances

EC2

Dashboard

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Successfully initiated termination (deletion) of i-0537e4e677258c7ec

Notifications 0 0 2 0 0 0

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☒	SJP_RIT_HSN1	i-0537e4e677258c7ec	Shutting-d...	t3.micro	3/3 checks pass	View alarms	eu-north-1b	-	-

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Home:

aws [Search] [Alt+S] Europe (Stockholm) SJP_RIT_HSN

EC2

Dashboard

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

▼ Images

AMIs

Resources

You are using the following Amazon EC2 resources in the Europe (Stockholm) Region:

Instances (running)	0	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	1
Key pairs	1	Load balancers	0	Placement groups	0
Security groups	2	Snapshots	0	Volumes	0

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance

Migrate a server

Service health

AWS Health Dashboard

Region

Europe (Stockholm)

Status

This service is operating normally.

EC2 Free Tier

Offers for all AWS Regions.

0 EC2 free tier offers in use

End of month forecast

0 offers forecasted to exceed free tier limit.

Exceeds free tier

0 offers exceeded and is now pay-as-you-go pricing.

View Global EC2 resources

View all AWS Free Tier offers

Account attributes

Default VPC



Experiment-2

Working with Amazon S3 Orchestrating Serverless Functions with AWS Step Functions

Working with Amazon S3 (Simple Storage Service) involves using it to store and manage data in the cloud. You can interact with S3 using the AWS Management Console, AWS CLI (Command Line Interface), or programmatically with SDKs such as Boto3 (for Python). Here's a basic overview of key concepts and how to get started.

Key Concepts

- Bucket: A container for storing objects (files) in S3.
- Object: A file stored in S3, identified by a unique key within a bucket.
- Region: AWS regions where your S3 buckets are located.
- Access Control: S3 offers various mechanisms (IAM policies, bucket policies, ACLs) to manage permissions.

Steps to Work with Amazon S3

1. Uploading File

Steps:

4. Open the [AWS Management Console](<https://aws.amazon.com/console/>) and log in.
5. Navigate to S3 by searching for it in the search bar and selecting S3.
6. In the S3 Dashboard, click on the bucket where you want to upload files.
7. In the bucket view, click the Upload button.
8. A window will appear where you can drag and drop files or entire folders from your computer.
9. Once the files are dragged into the upload window, click Upload to start uploading them to your S3 bucket.

Notes:

- Upload multiple files or even entire folders. S3 will maintain the folder structure if you upload a folder.
- The upload window shows the progress of the files being uploaded.

2. Moving or Copying Files Between Buckets

While S3 doesn't have a true "drag-and-drop" feature for moving files directly between buckets within the console, you can still copy or move files using the console in a similar way.

Steps:

1. Open the source bucket where the file is located.
2. Select the file(s) you want to move or copy.
3. Click on the Actions dropdown and select Copy or Move.
4. Go to the destination bucket.
5. Once inside the destination bucket, click Paste to copy or move the files.

While the drag-and-drop method isn't directly available for moving between buckets, this manual copy/move is a useful workaround.

3. Downloading Files

Downloading files from an S3 bucket can also be done directly in the S3 console.

Steps:



1. Go to the S3 Console and open the bucket containing the file you want to download.
2. Select the file you want to download.
3. Click on the file, and in the file details page, you'll see an option to Download.
4. The file will be downloaded to your local machine.

4. Deleting Files

Deleting files from S3 is straightforward via drag-and-drop actions in the console.

Steps:

1. Open the bucket in which the file is stored.
2. Select the file(s) you want to delete.
3. In the Actions dropdown, select Delete.
4. Confirm the deletion. The file will be removed from the bucket.

Tip: Be cautious with deletion, as it is a permanent operation unless versioning is enabled.

5. Organizing Files by Creating Folders

You can easily create new folders and move files into them via the S3 Console.

Steps:

1. In the S3 bucket view, click the Create Folder button.
2. Name the folder and click Save.
3. Once the folder is created, you can drag and drop files into that folder.

Note: Although folders are displayed as part of the UI, S3 uses "prefixes" to simulate a folder structure. The files aren't really in traditional folders but are organized by prefix names.

6. Changing Permissions Using the Console

Changing the permissions of files (objects) can also be done, although it doesn't involve drag-and-drop. You can adjust access control settings for your objects.

Steps:

1. Open the bucket and select the file for which you want to change permissions.
2. Click on the file to open its details page.
3. In the Permissions tab, you can set permissions using ACLs (Access Control Lists) or Bucket Policies.
 - You can make the file public or restrict access to specific users.

Tip: If you want to give public read access to files, ensure you adjust the public-read permissions.

7. Renaming Files

While S3 doesn't have a direct drag-and-drop method for renaming files, you can still easily rename a file by copying it and deleting the original.

Steps:

1. In the bucket, select the file you want to rename.
2. Click on Actions and select Copy.
3. Paste the file into the same bucket with the new name (change the name in the "Destination" path).
4. Once the file is copied with the new name, delete the original file.

8. Viewing and Editing Metadata

While S3 doesn't support direct drag-and-drop to modify metadata, you can view and edit metadata through the console:

Steps:

1. In the S3 console, select the file.
2. Click on the file to open its Properties tab.
3. Under Metadata, you can see and edit various metadata fields, such as content type and custom metadata.

Additional Features:



- Bulk Operations: Using drag-and-drop in S3 Console allows you to perform bulk uploads, download multiple files, or organize them by folders. This is particularly helpful when managing large sets of files.
- Transfer Acceleration: If you're uploading large files, consider enabling Transfer Acceleration for faster uploads to S3 using Amazon's global network.
- Versioning: If you have versioning enabled, you can easily revert to previous versions of a file, even if it's deleted or modified.